

Lu Wang

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AI Software Engineer

AI Software Engineer with 5 years of experience in backend engineering and LLM-based AI systems. Specialized in building production-grade AI systems with high-concurrency, low-latency architectures. Delivered production AI systems for large-scale e-commerce serving tens of millions of users across Europe, as well as enterprise AI agents for leading financial institutions. Experienced in Agent Runtime, Streaming, Tool Calling, Text2SQL, and personalization, with end-to-end ownership from system architecture to production deployment.

Work Experience

2025.08 - Present *AI Software Engineer*
Zalando Technology Development (Shenzhen) Co., Ltd.

Europe's largest online fashion and lifestyle platform. AI Software Engineer responsible for production AI systems powering the Zalando Assistant.

2021.04 - 2025.08 *Senior Software Engineer*
Thoughtworks Software Technologies (Shenzhen) Co., Ltd.

Global technology consulting company. Senior Software Engineer responsible for enterprise AI Agent and data analytics platform development for financial services.

Zalando Experience

Agent Runtime Upgrade and Real-Time Response Optimization

Led the upgrade of Zalando Assistant's core inference pipeline, including Product Tool integration, Streaming Runtime redesign, and OpenAI Responses API migration, building a production-grade Agent infrastructure supporting multi-tool execution.

Key Contributions

- **Designed and implemented dynamic Product Tool invocation**, enabling the Agent to retrieve real-time product information based on user intent and generate fact-grounded shopping responses.
- **Built a state-machine-driven Streaming Runtime to coordinate Tool Calling**, business events, and model outputs, eliminating duplicated responses and event-ordering issues in multi-tool execution. Reduced Suggestions API TTFT by ~25% while lowering token consumption.
- **Migrated the Agent Runtime** from Chat Completions to OpenAI Responses API, establishing a unified foundation for multi-model integration and multi-tool orchestration. Reduced end-to-end P95 TTFT by ~25%.

Personalization and Guided Shopping Capabilities

Led the personalization capability for Zalando Assistant, building a production recommendation system serving tens of millions of users across Europe. Optimized contextual understanding, user

memory, recommendation strategies, and system performance to address cold-start latency and scalability challenges in high-concurrency profile services.

Key Contributions

- **Built a multi-source personalization pipeline** combining behavioral history, conversation context, and user profiles. Designed a hierarchical memory model separating short-term conversation context from long-term user preferences, increasing recommendation entry engagement by 15%+.
- **Designed a hierarchical Redis cache** for user profiles with field-level invalidation, fault isolation, and degradation strategies, reducing profile-service traffic by 70%+ and lowering P99 latency by ~60% under high concurrency.
- **Designed an asynchronous warm-up framework** using Redis Registry and in-memory fallback, moving recommendation generation off the request path to background execution and reducing cold-start latency by ~60%.
- **Redesigned the recommendation strategy** using a multi-stage decision framework covering behavior recognition, recommendation planning, and fallback generation. Iteratively optimized the framework across 800+ real production scenarios, improving recommendation robustness beyond pure LLM generation.
- **Established a production evaluation framework** using Agent telemetry and live traffic. Implemented circuit breaking, gradual rollout, and degradation strategies to support continuous recommendation optimization with production reliability.

Product Comparison Skill

Designed and launched Product Comparison Skill for Zalando Assistant, enabling natural-language product comparison through a hybrid architecture combining Agent reasoning with deterministic code execution to deliver reliable and scalable comparison results.

Key Contributions

- **Built a conversation-level product state management mechanism**, supporting position-based, range-based, and follow-up references across multi-turn conversations. Increased engagement in product comparison scenarios by 20%+.
- **Decoupled Agent reasoning** from structured rendering, using the model for semantic understanding while deterministic code handled attribute extraction, price calculation, and table generation, significantly reducing frontend rendering failures.
- **Built deterministic validation pipelines** covering field parsing, business-rule validation, and fallback handling, eliminating missing fields, format drift, and rendering failures from pure LLM outputs while improving production stability.

Thoughtworks Experience

AI Agent Module for Pricing Management System

Built an enterprise Data Analysis Agent for a leading bank, enabling natural-language data analysis and policy Q&A for internal users across Text2SQL, knowledge retrieval, and analytical workflows.

Key Contributions

- **Designed and implemented** a multi-stage Agent workflow covering intent clarification, SQL generation, query execution, and result summarization with visualization, supporting complex queries and multi-turn clarification.
- **Continuously optimized reasoning workflows** and built a 1,000+ case evaluation suite, improving Agent accuracy by 20%+ while increasing complex query understanding and SQL generation stability.
- **Built a dual-layer retrieval mechanism** combining field- and value-level semantic search with reranking. Introduced SQL validation, exception retry, and execution safeguards to improve end-to-end pipeline reliability.

Pricing Management System: Access Control and Reporting

Module Owner

Led the design of an access-control module based on Hexagonal Architecture and DDD, built a Tablesaw-based in-memory reporting solution improving development efficiency by ~40%, and completed a zero-downtime database migration supporting millions of pricing requests per day.

After-Sales Management System

Module Owner & Core Backend Engineer

Led backend architecture and API integration for the after-sales module across headquarters, platform services, and WMS. Delivered the module on schedule, resolved 98% of integration blockers, and supported tens of thousands of daily spare-part transactions after launch.

Skills & Technologies

- **AI / LLM Application Development:** LLM Agent, Agent Workflow, PydanticAI, Tool Calling, OpenAI Responses API, Streaming, RAG, Text2SQL, Prompt Engineering, RAGAS
- **Retrieval and Vector Technologies:** FAISS, LlamaIndex, Embedding, Reranking, BGE-Reranker, pgvector
- **Programming and Backend:** Python, Java, FastAPI, Spring Cloud, Redis

Education

MSc in Computing, **Cardiff University**, UK
2019 – 2021

BA in Marketing, **Shanghai University of Engineering Science**, China
2015 – 2019

汪露

AI软件工程师

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5年后端/LLM Agent研发经验, 专注于生产级AI系统与高并发、低延迟服务架构设计。具备欧洲千万级 C 端智能导购、头部银行企业级Agent两类生产环境落地经验。擅长构建Agent Runtime、Streaming、Tool Calling、Text2SQL 与个性化推荐系统, 主导多项核心能力从架构设计、工程落地到线上优化, 持续提升系统稳定性与可扩展性, 具备丰富的海外跨团队协作经验。

工作经历

2025.08-至今 **AI软件工程师** **嘉兰朵(深圳)科技发展有限公司**

欧洲第一时尚电商平台。担任 AI 软件工程师, 负责Zalando Assistant 核心能力建设。

2021.04-2025.08 **高级软件工程师** **思特沃克软件技术(深圳)有限公司**

全球领先的软件咨询公司。担任高级软件工程师, 负责企业级Agent与数据分析平台建设。

嘉兰朵/Zalando项目经验

个性化推荐与导购能力建设

主导 Zalando Assistant 个性化导购能力建设, 为千万级欧洲用户打造智能推荐系统, 围绕上下文感知、用户画像、推荐策略及系统性能持续优化, 解决冷启动延迟高、服务高并发等核心问题。

核心职责:

- **个性化输入链路构建:** 构建“历史行为 + 对话上下文 + 用户画像”的个性化输入链路, 解决原推荐逻辑仅依赖静态行为数据、缺乏对话连续性感知的问题, 提升推荐相关性与新特性曝光率, 推动推荐入口互动率提升**15%+**。
- **高并发画像缓存设计:** 在不改动原有业务调用链路的前提下, 为下游用户画像服务增加缓存层, 实现 TTL 缓存、字段级失效、异常隔离和空值过滤能力, 使下游用户画像服务调用量减少**70%+**, 高并发场景下接口 P99 延迟下降约**60%**。
- **异步Warm-Up架构设计:** 设计**Redis Registry + 本地内存回退**的异步预计算框架, 将耗时推荐生成从同步请求链路迁移至异步后台执行, 实现主链路解耦, 首屏冷启动耗时下降**~60%**。

- **推荐决策框架设计与落地：**主导商品详情页推荐策略重构，设计"行为识别→推荐方向决策→生成兜底"多阶段决策框架，结合**800+**真实线上场景持续迭代，解决纯生成不可控、静态推荐模板个性化不足等问题，保障不同用户与商品场景下均可稳定产出高相关推荐问题。
- **生产稳定性治理：**基于Agent埋点与线上真实流量建立策略评估闭环，设计工具调用熔断、灰度发布及异常降级机制，实现推荐策略持续迭代与生产环境稳定演进。

Agent Runtime 主链路升级与实时响应优化

主导Zalando Assistant核心链路升级，完成商详情 Tool 接入、Streaming 重构及 OpenAI Responses API 迁移，构建支持多工具调用的生产级 Agent 基础设施。

核心职责：

- **商品详情 Tool 接入与 Agent 编排：**设计并实现商品详情 Tool 动态调用能力，Agent 可根据用户对话意图拉取实时商品信息补充回答，弥补纯模型输出事实缺失问题。
- **Streaming 响应架构设计与延迟优化：**搭建状态机驱动的 Streaming Runtime，统一 Tool Calling、业务事件及模型输出处理，解决多工具调用场景下输出重复及时序错乱问题；Suggestions API TTFT 降低~**25%**，同时减少 Token 消耗，降低线上推理成本。
- **OpenAI Responses API 迁移与扩展能力建设：**完成 Agent 主链路从 ChatCompletions 向 Responses API 的迁移，统一 Tool Calling 与 Streaming 处理链路，为未来多模型接入及多工具并行调度提供统一基础设施，链路 P95 TTFT降低~**25%**。

Zalando Assistant 商品对比Skill能力研发

设计并上线 Product Comparison Skill，支持自然语言多商品比较与决策，通过 Agent 推理与确定性代码执行解耦，构建稳定可扩展的商品比较能力。

核心职责：

- **多轮商品状态管理：**构建对话级商品对象管理机制，支持序号、区间及追加引用解析，持续维护比较对象状态，商品比较场景互动率提升~**20%**。
- **LLM 与代码职责解耦：**拆分Agent推理与结构化渲染职责，模型负责语义理解，代码统一处理属性提取、价格计算与表格生成，显著降低前端渲染异常。
- **结构化输出稳定性治理：**建立字段解析、业务校验、异常回退等确定性处理流程，解决纯 LLM 输出存在的字段缺失、格式漂移及渲染失败问题，提升生产环境稳定性。

思特沃克/Thoughtworks项目经验

定价管理系统AI Agent模块 国内知名银行

该项目基于 AIGC 技术，服务于银行内部员工，目标是提升非技术人员的数据分析与政策查询能力。核心功能包括：通过 **Text2SQL** 实现结构化数据的自然语言查询，以及基于知识库

的政策类问答。作为核心开发工程师，负责数据分析模块的整体设计与落地，包括数据分析链路构建、SQL 生成优化与系统联调上线。

核心职责：

- **Agent 应用开发：**主导Agent应用开发，设计并实现“意图澄清 → SQL生成 → 查询执行 → 结果总结与可视化”的多阶段 Agent 工作流，支持复杂查询、多轮澄清和分析结果自动解读。
- **Prompt 工程与效果评估：**针对复杂业务查询场景持续迭代 Prompt 模板与推理流程，构建 1000+ 条真实业务测试集用于效果评估和回归验证，提升模型端到端问答准确**20%+**，增强模型对复杂查询意图的理解能力和 SQL 生成稳定性。
- **SQL 生成优化：**梳理高频业务语料，并基于向量数据库构建字段+值双层召回的 Rerank 机制；引入 SQL 校验、异常重试等机制，增强整体链路稳定性。

RAG智能问答Chatbot - 核心开发工程师

基于 LlamaIndex + FAISS 构建文档切片、向量化、召回与答案生成链路，并引入 RAGAS 评估答案相关性和上下文覆盖率，形成可复用的 RAG 研发与评估模板。

定价管理系统（权限与报表）-模块负责人

主导基于六边形架构+DDD的权限模块设计，自研内存数据加工方案（Tablesaw），报表开发效率提升约40%；推动MySQL到新架构的数据库迁移，实现0业务感知、每日数百万笔定价请求稳定运行。

售后管理系统 – 模块负责人 & 核心后端工程师

负责售后模块后端架构与API对接，协调总部、中台、WMS多系统集成；所负责模块准时交付，联调阻塞修复率~**98%**，系统上线后支撑日均数万级备件交易流转。

技能

- **AI & LLM Agent开发：**LLM Agent、Agent Workflow、PydanticAI、Tool Calling、OpenAI Responses API、Streaming、RAG、Text2SQL、Prompt Engineering、RAGAS
- **检索与向量技术：**FAISS、LlamaIndex、Embedding、Rerank（BGE-Reranker）、pgvector
- **编程&后端：**Python、Java、FastAPI、Spring Cloud、Redis

教育背景

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| • 英国卡迪夫大学 | Computing | 2019-2021 | 硕士 |
| • 上海工程技术大学 | 市场营销 | 2015-2019 | 本科 |